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Gas Sensor Node: An Innovative Scalable Solution for Continuous Monitoring and Quantification of Methane Fugitives

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Abstract

Methane fugitive emissions represent a critical challenge for the oil and gas industry, with implications for safety, environmental sustainability, and regulatory compliance. This paper presents a novel, scalable solution for continuous, source-level monitoring and quantification of methane leaks, aligned with emerging frameworks such as OGMP 2.0. The system consists of autonomous sensor nodes deployed directly on potential emission points within plant infrastructure. Each node integrates a methane-sensitive NDIR sensor, environmental probes, LoRa connectivity, and photovoltaic energy supply, housed in an ATEX Zone 0-certified enclosure. Three pilot campaigns—two in onshore facilities and one offshore—have demonstrated detection sensitivity below 1 gram per hour, confirming the system's reliability and robustness. All observed emissions, ranging from 1 to 100 g/h, were validated via conventional LDAR techniques. Data is wirelessly transmitted to a gateway and visualized through a cloud dashboard for post-processing and maintenance planning. This architecture supports digital transformation in emissions management by enabling persistent, low-impact monitoring with minimal manual intervention, contributing to Net Zero targets and operational excellence.

Introduction

Methane fugitive emissions pose a critical challenge for the oil and gas industry. Methane (CH_4), the main component of natural gas, is a highly potent greenhouse gas with a Global Warming Potential (GWP) approcimately82.5 times that of CO_2, over a 20-year horizon (Forster et al., 2021). Even relatively small unintentional leaks from industrial facilities, known as fugitive emissions, are increasingly recognized as a key target for mitigation. Controlling these emissions is therefore essential for improving environmental performance (reducing greenhouse gas emissions), ensuring process safety (preventing fire/explosion risks), and achieving compliance with increasingly stringent regulations.

Conventional approaches to methane mitigation rely on periodic Leak Detection and Repair (LDAR) programs, where technicians perform routine surveys using handled detectors or Optical Gas Imaging (OGI) cameras to locate and address leaks (Fox et al., 2020). While widespread and regulatory-compliant (Allen, 2014), these traditional methods have significant limitations:

- Time gaps in emission surveillance: LDAR surveys are typically conducted on monthly, or quarterly intervals, or even less frequently. Large leaks that occur between inspections often go undetected for extended periods, leading to unreported emissions. Studies show that methane emissions follow a skewed distribution (Brandt et al., 2016), where fewer than 5% of leaks account over 50% of total emissions – so called "super emitters" – many of which are intermittent and short-lived (Cusworth et al., 2023, Xia et al., 2024).
- Manual limitations and safety constraints: LDAR surveys often require physical access to various parts of the facility, including hard-to-reach components such as elevated equipment in hazardous areas. This not only increases operational cost and effort, but also raises risks for personnel, particularly in environments with dangerous conditions (toxic or explosive gas accumulations, extreme weather, etc.). (Trinity Consultants, 2019). Even with OGI technology, an operator must be within line-of-sight of the leak. Thus certain zones remain inaccessible unless equipment is shut down, delaying detection.
- Lack of continuous quantification: While OGI cameras enable visual detection of gas plumes, they typically do not provide real-time quantification of the leak rates (Fox et al., 2019). In most cases, operators estimate the emission magnitude or rely on secondary instruments during the inspection (Johnson et al., 2021). Moreover, because methane emissions often fluctuate over time, a single measurement may not reflect the actual average emission rate. This limitation complicates accurate inventorying of total emissions and impedes data-driven prioritization of leak repairs.

Meanwhile, regulatory and voluntary frameworks for methane emissions management are becoming increasingly stringent. The Oil and Gas Methane Partnership 2.0 (OGMP 2.0), led by the United Nations Environment Programme (UNEP), is a global initiative promoting high-accuracy methane emissions reporting through a structured, multi-tier system. Level 4 (L4) reporting requires direct, source-level measurements of emissions at individual leak points, while Level 5 (L5) introduces site-level quantification using aerial, drone-based, or satellite observations, with reconciliation against aggregated L4 data. Companies that achieve full L4 and L5 data integration are recognized as OGMP "Gold Standard" reporters, demonstrating that most of their emissions are empirically measured rather than estimated (UNEP, 2020).

In parallel, emerging regulations — such as the European Union Methane Regulation — are introducing mandatory monitoring requirements and are designed to align with OGMP 2.0 principles. These may include higher inspection frequencies and, in some cases, continuous monitoring obligations (European Commission, 2024). Consequently, operators are under growing pressure to adopt advanced monitoring technologies that enable both persistent emissions tracking and robust quantification at the source and facility level (IEA, 2022).

In response to these challenges, this paper introduces the Gas Sensor Node (GSN) system – an innovative, scalable, and field-deployable solution for continuous methane leak detection and quantification at the source level. The GSN network is designed to continuously monitor potential emission points across an industrial site, automatically detecting leaks and measuring their flow rates in near real-time.

The key objectives of the system are to:

a. generate persistent, source-level (L4) methane data across every emission point in a facility,
b. operate as a full energy-autonomous solution, making it suitable for remote or otherwise inaccessible locations,
c. provide real-time notification and actionable analytics to support proactive maintenance and early intervention – well ahead of conventional LDAR timelines – and to ensure alignment with Net Zero emissions goals.

A distributed network of GSNs enables a shift from occasional surveys to real-time, continuous monitoring – enabling faster leak detection and stronger methane control across the facility while driving operational excellence.

Technology Description

Gas Sensor Node (GSN) is a distributed hardware and software architecture for high-resolution continuous methane monitoring at source level. It consists of multiple autonomous sensor units (the "nodes") deployed across an industrial site and a centralized data processing and integration platform.

Each node is a self-contained, intrinsically safe unit capable of measuring methane at its location, estimating the leak rate, and wirelessly transmitting the data.

The following section describes the architecture of a GSN node, the installation and deployment strategy, and the end-to-end data pipeline – from field-level detection to actionable user insights.

System Architecture

Each Gas Sensor Node is engineered to ensure robustness in harsh field conditions, compliance with intrinsic safety standards, and complete operational autonomy. Its design enables reliable, long-term deployment even in hazardous environments, without need for manual servicing or external power.

These nodes operate as part of a distributed sensing network (Fig. 1), continuously monitoring potential emission points across a facility. Their internal architecture (Fig. 2) is optimized for ultra-low power consumption, autonomous energy harvesting, and accurate environmental sensing.

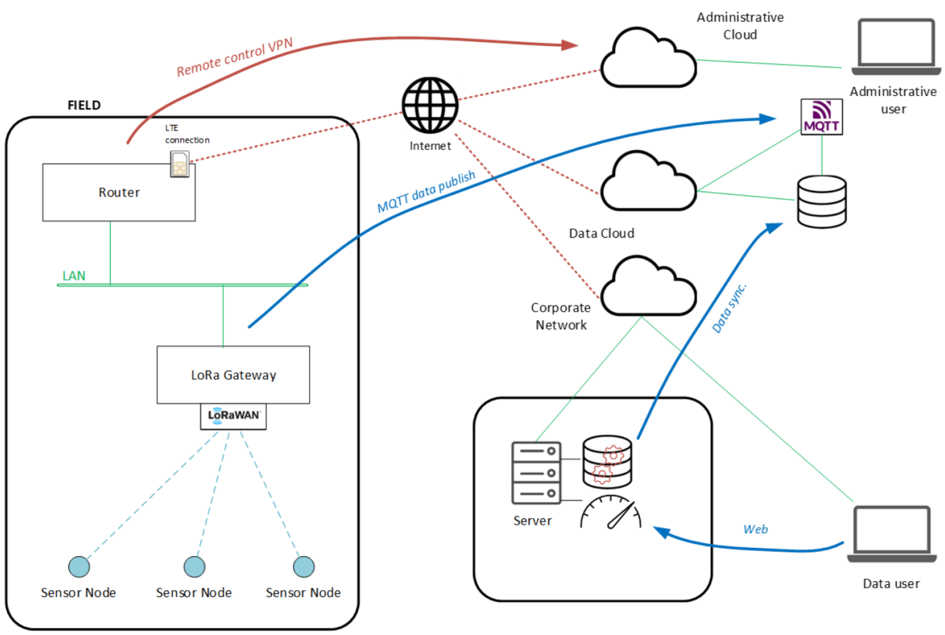

Figure 1—Gas Sensor Node system's architecture

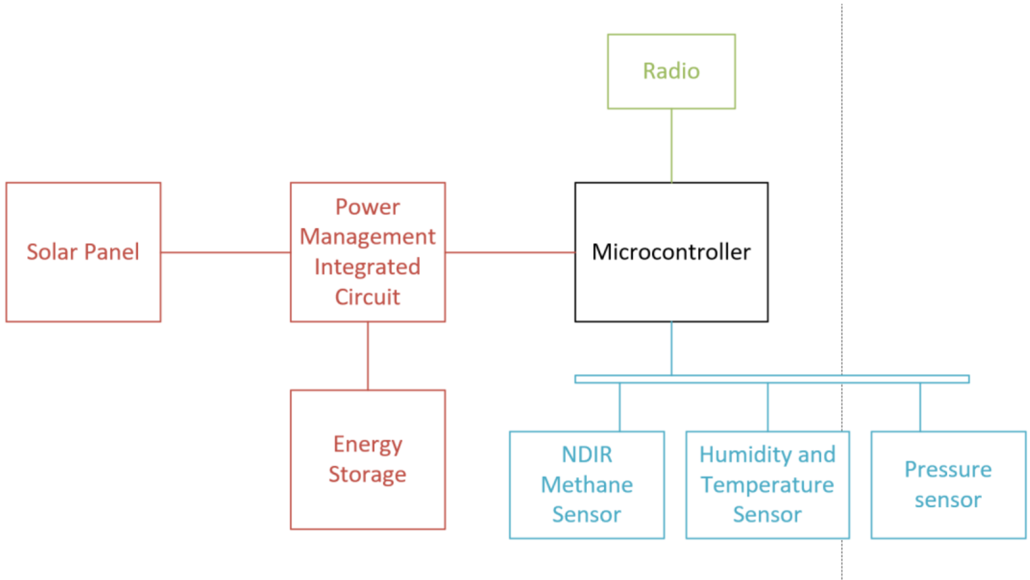

Figure 2—Block diagram of a Gas Sensor Node

The key components and functional features of a node are the following:

- ATEX Zone 0 certified assembly: The entire sensor and electronics are compliant with ATEX Zone 0 requirements. This allows the node to be installed directly in areas with continuous methane presence (e.g., directly on a leak-prone flange) without ignition risk. The circuitry is intrinsically safe, operating at ultra-low energies, and integrates safety barriers that eliminate the risk of sparks or overheating. This certification is essential for placing sensors inside hazard zones rather than at a safe distance, thereby enhancing leak detection sensitivity and accuracy (European Commission, 2024).
- Methane detection via NDIR sensor: Each node integrates a Non-Dispersive Infrared (NDIR) methane sensor that continuously samples ambient air, providing measurements of methane concentration (expressed in ppm or % volume). This optical sensing method offers high selectivity to methane, as it detects only gases absorbing infrared at methane-specific wavelengths, thereby minimizing false positives from other compounds. The sensor exhibits low calibration drift, maintaining accuracy over long deployment periods without frequent recalibration – a key feature for continuous monitoring applications. Thanks to the combination of the containment setup and the proprietary quantification algorithm, the system can detect very small leaks—corresponding to methane concentrations above 1000 ppm and leak rates below 1 gram per hour under standard environmental conditions—despite the moderate intrinsic sensitivity of the sensor—enabled by the integration of containment with the proprietary quantification algorithms. This level of sensitivity enables the detection of early-stage emissions, aligning with the project objective of proactive leak identification before escalation.
- Environmental sensors: Each node includes environmental sensors for temperature, pressure and humidity to refine the methane readings under varying conditions. For instance, temperature and pressure affect infrared absorption by altering gas density and source intensity, while humidity can interfere with infrared measurements due to water vapor absorption in certain bands. The node's firmware applies correction algorithms based on these values, maintaining data accuracy.
- Ultra-low-power node: Each node's hardware and firmware is designed for ultra-low power consumption and supports over-the-air updates. Its main functions include: periodically acquiring data from the methane and environmental sensors, processing measurements, and preparing data

packets for transmission. To conserve energy, the firmware puts the system into deep sleep between measurements cycles. Minimum duty cycle involves waking every 5 minutes and is fully configurable depending on site need and seasonal light condition.

- Wireless communication: Each node communicates data using a LoRa-based radio module. The physical layer leverages LoRa modulation to ensure long-range, low-bandwidth transmission, suitable for reaching gateways positioned several kilometres away – even in obstructed or metallic environments typical of industrial sites. Data packets include the node ID, timestamp, methane concentration (or estimated leak rate), power status, and notification flags. These are short uplinks transmitted at scheduled intervals to minimize energy consumption. Communication is managed via the LoRaWAN protocol stack (LoRa Alliance, 2020), which enables bidirectional, asynchronous data exchange: in addition to transmitting sensor data, nodes can receive configuration updates (e.g. sampling interval changes or firmware commands). This bidirectionality is essential to support adaptive behaviour and remote maintenance, while preserving ultra-low power consumption. LoRaWAN ensures reliable data transfer through built-in acknowledgment and retransmission mechanism, while the physical layer employs cyclic redundancy checks (CRC) to detect transmission errors. In typical industrial applications, this architecture allows large sites to be covered with a single gateway acting as a communication hub, significantly reducing infrastructure complexity in brownfield deployments and allowing gateway placement in non-hazardous zones. The system operates in the ISM band (e.g., 868 MHz in Europe, 915 MHz in North America).
- Cloud architecture and cybersecurity considerations: The data infrastructure supporting the Gas Sensor Node system is built upon a distributed cloud-based architecture, designed to ensure high availability, scalability, and robust cybersecurity. Sensor data and control commands are transmitted via secure protocols to redundant cloud services, which manage storage, processing, and user access. The architecture ensures role-based access control, encryption of data at rest and in transit, and authentication mechanisms aligned with enterprise-level cybersecurity policies. These features are essential for protecting sensitive operational data in industrial environments and preventing unauthorized access to critical methane monitoring systems. Moreover, the architecture separates the presentation layer from core infrastructure components, reducing the risk of application-level vulnerabilities while enabling integration with corporate platforms and regulatory reporting systems.
- Energy harvesting power supply: A standout feature of the GSN design is that each node operates autonomously using a self-contained solar energy system – no wired power or battery replacement is required. A compact photovoltaic panel on the enclosure charges internal high-capacitance supercapacitor, which stores sufficient energy to sustain operation during the night and low-light periods. The electronics are optimized for ultra-low power consumption, and the supercapacitor was selected over traditional batteries due to its ability to endure hundreds of thousands of charging cycles without degradation and to function reliably across a wide temperature range (-40°C to +60°C), as commonly encountered in outdoor industrial environments. This design enables multi-year, maintenance-free operation without battery replacements – a key advantage for safety and reliability in hazardous locations.

In summary, the Gas Sensor Node is an intrinsically safe, solar-powered methane monitor designed for continuous, unattended operation. It can be installed directly within hazardous areas across the plant, measuring methane concentration at source-point and transmitting data wirelessly. The system requires no wired power, no routine maintenance, and no operator intervention, making it ideally suited for scalable, long-term deployment in industrial environments.

Deployment Strategy and Sensors Mounting

One of the key design priorities of the GSN system is ease of installation on existing plant equipment. The objective was to enable rapid, non-intrusive deployment – allowing hundreds of sensor nodes to be installed across a live facility without requiring modifications or process interruptions. To achieve this, a flexible flange-mount system was developed, enabling each node to be positioned directly at the source of potential emissions.

- Universal flange adapter: In oil and gas facilities, many methane leaks originate at mechanical interfaces – such as flanged joints, valve packings, pump seals, etc. To address this, we developed a universal flange adapter: a flexible, form-fitting cover that clamps onto standard flange geometries and creates a semi-sealed "dome" around the interface. The Gas Sensor Node is integrated directly into this dome, acting as its top cap. The adapter is made of a chemically resistant plastic foil, which provides mechanical stability while accommodating small variations in flange dimensions. Depending on the installation scenario, it can be secured either via existing flange bolts or with adjustable straps. This mounting approach creates a localized enclosure around the flange gap, forming a defined control volume. In case of a leak, this allows for accurate source-level measurement of the emission, rather than relying on dispersed ambient air readings.
- Transparent window for solar: The flange cover incorporates a transparent window positioned directly above the node's photovoltaic panel. This window, made of UV-resistant polycarbonate, allows sunlight to reach the panel even when the node is installed on the side or underside of horizontal piping. By enabling light transmission while protecting the rest of the device, the window ensures continuous solar charging under a wide range of installation angles. Field tests confirmed that this configuration provides adequate energy input to maintain capacitor charge even under indirect daylight conditions, thanks to the node's ultra-low power requirement.
- Versatility for other sources: To address a broader range of emission points, the mounting system includes a variety of adaptable fixtures – such as valve stem collars that secure the sensor near packing gland, or a magnetic bases that allow placement on steel surface adjacent to a weld or fittings suspected to leak. In one pilot deployment, custom clamps were used to install sensors around compressor seals. This modular approach ensures that any location identified as a potential methane emission source can be equipped with a GSN node in minutes, without requiring specialized tools or equipment modifications.
- Non-intrusive installation: A key advantage of the GSN system is that installation does not require process shutdowns or mechanical modification. Thanks to the flexible and adaptive mounting solutions, sensor can be deployed while the plant remains in operation ("hot installation"), as they simply enclose components externally without interfering with their function or integrity. This enables continuous leak monitoring to begin immediately, without waiting for scheduled turnarounds or maintenance windows.
- Deployment procedure: A typical deployment begins with a review of the facility's P&IDs and historical LDAR records to identify high-risk emission points – such as aging valves, known chronic leakers, or high-pressure interfaces. Field teams then install GSN nodes at those locations. Upon activation, each node joins the LoRaWAN network and is associated – via mobile application – with its physical location and corresponding equipment tag. Within a few hours of installation, each sensor establishes a baseline reading: some may immediately detect elevated concentrations, while others confirm integrity. From that point, the system operates autonomously, maintaining continuous 24/7 surveillance of all monitored points. A visual example of the installation process and full-scale deployment is provided in Fig. 3.

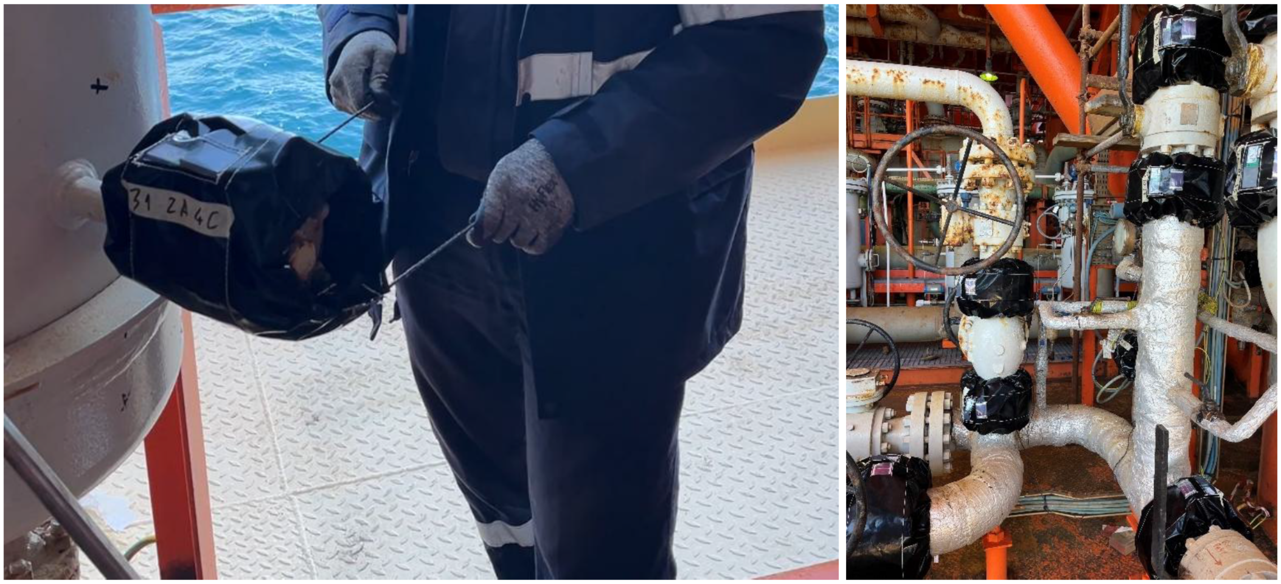

Figure 3—Gas Sensor Node deployment: (left) installation of the sensor on a flange; (right) multiple nodes mounted on piping interfaces within an operational facility.

This straightforward installation process makes the GSN system highly scalable: dozens to hundreds of autonomous sensors can be deployed across a site's emission sources. Such large-scale coverage would be impractical using wired instrumentation or frequent manual inspections, but becomes feasible through compact, wireless, self-powered nodes.

Data Acquisition and Processing Pipeline

Once deployed, the GSN network continuously acquires data on methane concentrations and estimated leak rates at each sensor node. The end-to-end data pipeline comprises four key stages:

Stage 1. Local Sensing and Leak Detection

Each GSN node monitors methane concentration at its location at a fixed sampling interval, typically configurable between 5 minutes and several hours, depending on desired monitoring resolution and available solar energy. A built-in algorithm filters out noise and computes an average concentration over a short acquisition window (several seconds) to reduce fluctuations. This processed value is then compared to a reference baseline – established during initial deployment – and to predefined thresholds. If the concentration exceeds the lower threshold, the node flags the reading as a potential leak. Crucially, the node can dynamically increase its sampling frequency in response to sudden spikes; for instance, if methane levels rise sharply, it may temporarily switch to high-frequency acquisition (e.g., every 30 seconds) to capture the event in greater detail.

Stage 2. Wireless Data Transmission

After each measurement, the node's microcontroller packages the data and transmits it via LoRa radio to a central gateway. The LoRaWAN network is configured such that all nodes report to one or more gateways strategically positioned across the site (e.g., on a communications tower or within a control room). Thanks to LoRa's long-range and obstacle-penetrating capability, a single gateway can typically provide coverage of 1 to 2 km in obstructed industrial environments, and up to 10-15 km in open-air, line-of-sight conditions (LoRa Alliance, 2020). In field deployment of GSN, a single gateway successfully covered an entire onshore gas plant. In offshore scenario, LoRa communication between platforms separated by up to 5 km was also achieved.

Stage 3. Data Aggregation and Cloud Processing

The LoRa gateway acts as a bridge, collecting packets from all sensor nodes and forwarding them – typically via cellular, Ethernet or Satellite connection – to a cloud-based platform. There, the central GSN platform aggregates the incoming data streams and applies the proprietary quantification algorithm to convert raw methane concentration readings into estimated leak rates for each source (e.g., in grams per hour or milliliters per minute). This conversion relies on calibration models derived from controlled release experiments. The proprietary quantification algorithm currently achieves a leak rate accuracy of 65%, and its effectiveness has been proven with methane flow rates between 0.3 and 5000 mL/min. For each sensor installation, the system accounts for the specific geometry of the measurement volume (e.g., a flange enclosure of known size) and correlates the observed concentration to a corresponding leak rate, while factoring in parameters such as flange diameter and wind intensity. For instance, a reading of 10% vol inside a containment around a 3" flange corresponds to a leak of approximately 0.03 L/min. Regarding the wind contribution, it has been verified that, above 1 m/s, this always has the effect of diluting the concentration of methane within the containment. To remove the wind factor from the data, the procedure involves collecting methane concentration data over a period of a few days. Each concentration is then converted into its flow rate according to the relationship described in the proprietary quantification algorithm. If the flow rate of the leakage remains constant during this time interval and that variations are solely induced by the wind, the 90th percentile of the leak flow value is considered. This makes it possible to consider only situations in which wind intensity is minimal and does not affect quantification, while eliminating insignificant spikes. All data is timestamped and stored in a structured database, enabling the creation of a continuous, source-level emissions log.

Stage 4. User Dashboard and Notification System

End-users – such as operators or environmental engineers – interact with the GSN system through a web-based dashboard and notification system. The dashboard provides a real-time map of the facility, displaying each sensor location with color-coded markers: green for normal readings, yellow for elevated readings, and red for leaks. By selecting a marker, users can access time-series graphs of methane concentration and leak rate. The system can automatically trigger notification – via email or SMS notification, or with SCADA system integration – when a leak exceeds predefined thresholds (e.g., >50 mL/min), ensuring that maintenance teams are notified promptly. The dashboard also enables advanced analytics, such as identifying chronically leaking components for prioritization or quantifying the impact of corrective actions by visualizing before-and-after leak rates to assess repair effectiveness. All measurement data is stored in the cloud, supporting methane emissions reporting at various aggregation levels (daily, monthly, annual) in line with regulatory or voluntary reporting frameworks.

To estimate methane leak rates from measured concentrations, the GSN system uses a proprietary algorithm calibrated through controlled release tests. These tests established reference correlations between gas concentration and emission flow under known conditions. The algorithm also incorporates physical models and compensates for environmental factors such as airflow and dispersion. This enables the system to translate concentration readings (ppm) into approximate leak rates, supporting source-level quantification with minimal user intervention.

Field Implementation and Results

To validate its performance under real-world conditions, the Gas Sensor Node system has been deployed across multiple pilot sites. As of mid-2025, three field implementations are ongoing: two at onshore natural gas treatment facilities and one at an offshore production platform. These pilots have enabled evaluation of the system performance and effectiveness across different operational context – ranging from compact to complex infrastructures, and from temperate inland locations to marine environments.

Multi-Site Implementation Pilot Summary

Between May and June 2025, the Gas Sensor Node (GSN) system was deployed across two onshore gas processing facilities and an offshore gas production platform as a field pilot, with a combined total of 314 sensors installed (120 offshore and respectively 98 and 96 at the two onshore sites). The monitoring periods spanned roughly 2–3 weeks at each location, during which the network operated continuously with ~97% system uptime and ~100% gateway uptime. This high reliability meant data was captured nearly continuously at both sites. About 80% of the sensors reported at least one measurement every day – the remainder, located in very shaded areas, still logged data intermittently and were later adjusted to improve solar charging. Overall, these deployments confirmed the GSN's ability to run autonomously in real industrial conditions, maintaining robust wireless connectivity and power over an extended period.

In most areas, the total detected methane leak rate was essentially zero. Nearly all sensors recorded zero or near-zero readings throughout the campaign, indicating tight equipment. The absence of further notification gave operators confidence that no hidden issues were present, reinforcing the effectiveness of existing LDAR (Leak Detection and Repair) programs and suggesting that manual survey intervals could potentially be optimized.

Three significant leaks were recorded during the campaign and promptly identified within the first few days of deployment. Combined, these events accounted for a total emission rate of ~0.0876 kg/h (87.6 g/h or 2.04 L/min), equivalent to roughly 0.77 tons of methane annually if left unaddressed – a non-trivial quantity in both environmental and economic terms. Notably, over 90% of the emissions came from just three sources, confirming the typical distribution often observed in gas facilities:

- Leak A (largest source): A GSN on the fuel gas line feeding a gas heater registered methane concentrations corresponding to 315 ±205 of methane (~0.013 kg/h).
- Leak B (second-largest source): Another sensor the fuel gas line feeding a glycol reboiler found a leak 279±167 mL/min (about 0.011 kg/h). After maintenance, the leak rate spiked to ~ 2494 mL/min (nearly 0.1kg/h)
- Leak C (third source): A smaller but still significant leak (~25 mL/min, or 0.001 kg/h) was detected on the same fuel gas line feeding the gas heater as Leak A (but at a different flange/connection point).

Fig. 4 and Fig. 5 illustrate time-series data from sensors showing continuous leak rates under stable environmental conditions, confirming the robustness of the monitoring and the reliability of the system's detections.

This scenario validated the common phenomenon of major emission contribution due to few biggest leaks and demonstrated the value of continuous monitoring – the GSN system notified plant personnel within hours of installation to the biggest leaks.

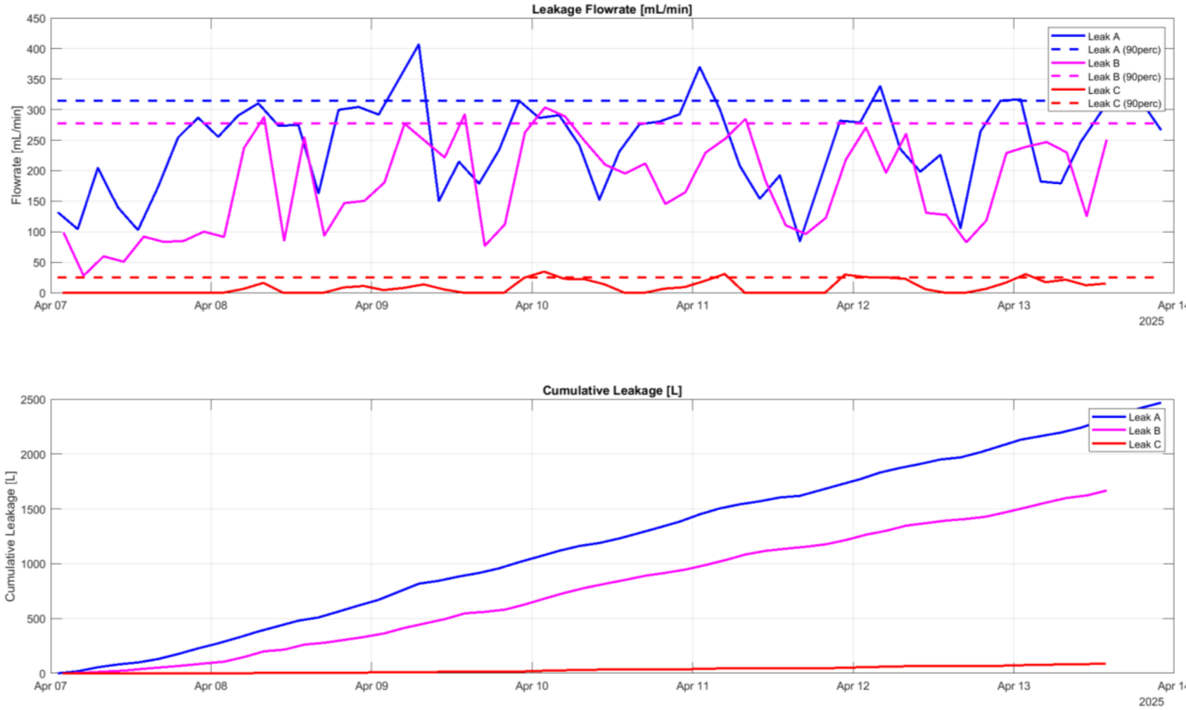

Figure 4—Time series data from two GSN devices monitoring the leakage events. The upper plot shows estimated flowrates over time, while the lower plot displays cumulative leakage volumes starting from the beginning of the observation window (April 7), not from the initial leak occurrence.

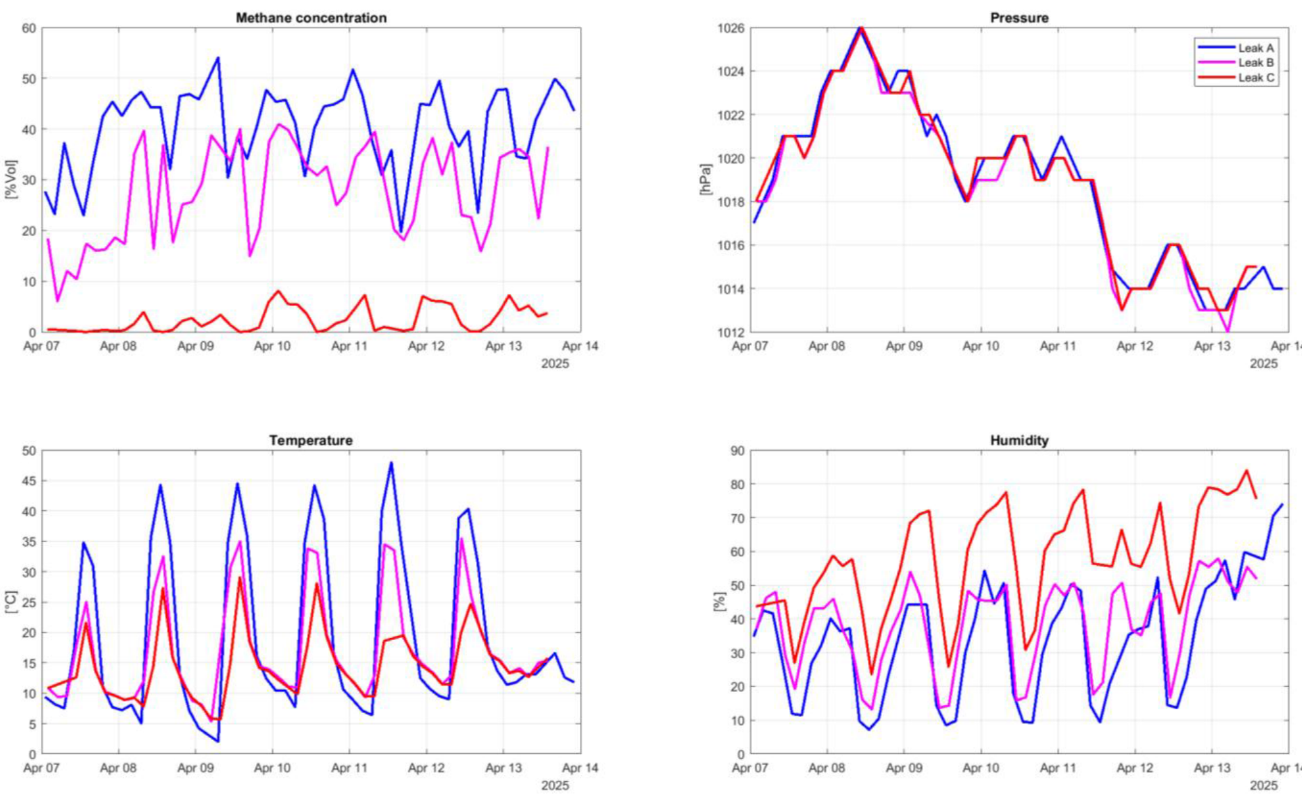

Figure 5—Monitoring data from GSN sensors showing the gas concentration (top left) associated with the leakage events, while environmental parameters—temperature (top right), pressure (bottom left), and humidity (bottom right)—remain within normal fluctuations. This confirms that the observed anomaly is leak-related and not driven by external conditions.

Upon receiving the GSN notification, the site operations team performed targeted inspections to verify the sensor readings. Using both FID sniffer and Hi-Flow Sampler (HFS), they confirmed methane plumes at the exact locations GSN had identified. Furthermore, HFS instrument was used to quantify Leak A's rate, measuring 235±44 ml/min -compliant with the GSN's estimate, which affirmed the accuracy of the sensor quantification. Leaks B and C was likewise confirmed with portable instruments. This cross-check gave the team confidence in the new continuous monitoring system. It also highlighted an advantage of GSN's 24/7 coverage: one of the leaks (Leak B) was intermittent, due to periodic maintenance of the reboiler unit, and likely would have been missed if a single manual survey happened during a non-leaking interval. Instead, the continuous data captured those variations, ensuring the issue was not overlooked. In addition, the HFS measurements confirmed the magnitude of the leaks, with flow rates closely matching the GSN estimates —as detailed in Tab. 1. This further validated the system's quantification accuracy across multiple sources and reinforced its value as a reliable tool for continuous emission monitoring.

Table 1—Direct verification of GSN records on-site using a Hi-Flow Sampler (HFS). The table compares the GSN-flow estimation values with spot HFS measurements. Results show good consistency in both magnitude and detection capability, supporting the reliability of the GSN system for quantification of component-level emissions.

Leak	Process	Source Item	GSN Flow Rate [mL/min]	HFS Measured Flow Rate [mL/min]
Leak A	Gas Heater	Flange	315 ±205	235 ± 44
Leak B	Glicol Reboiler	Flange	279 ±167	176± 65
Leak C	Gas Heater	Flange	25 ±16	13 ± 9

Armed with reliable, real-time data pinpointing each leak, operators moved quickly to fix the problems. Maintenance was immediately scheduled for the identified sources: the flange associated with Leak A was tightened and its gasket replaced, and the flange causing Leak B was serviced as well. The outcomes were clearly visible in the GSN data – after repairs, the corresponding sensor readings dropped, confirming that the leaks were successfully stopped. In one instance, Leak B, the system even provided feedback on maintenance quality: right after a flange repair, the sensor showed a sudden methane concentration increase at that source, prompting the field operators to reinspect and adjust the fix (they discovered the gasket hadn't seated correctly on the first attempt). This real-time feedback loop, only possible with continuous monitoring, helped ensure that issues were truly resolved before declaring the all-clear. Fig. 6 illustrates the sensor trend before and after the maintenance intervention.

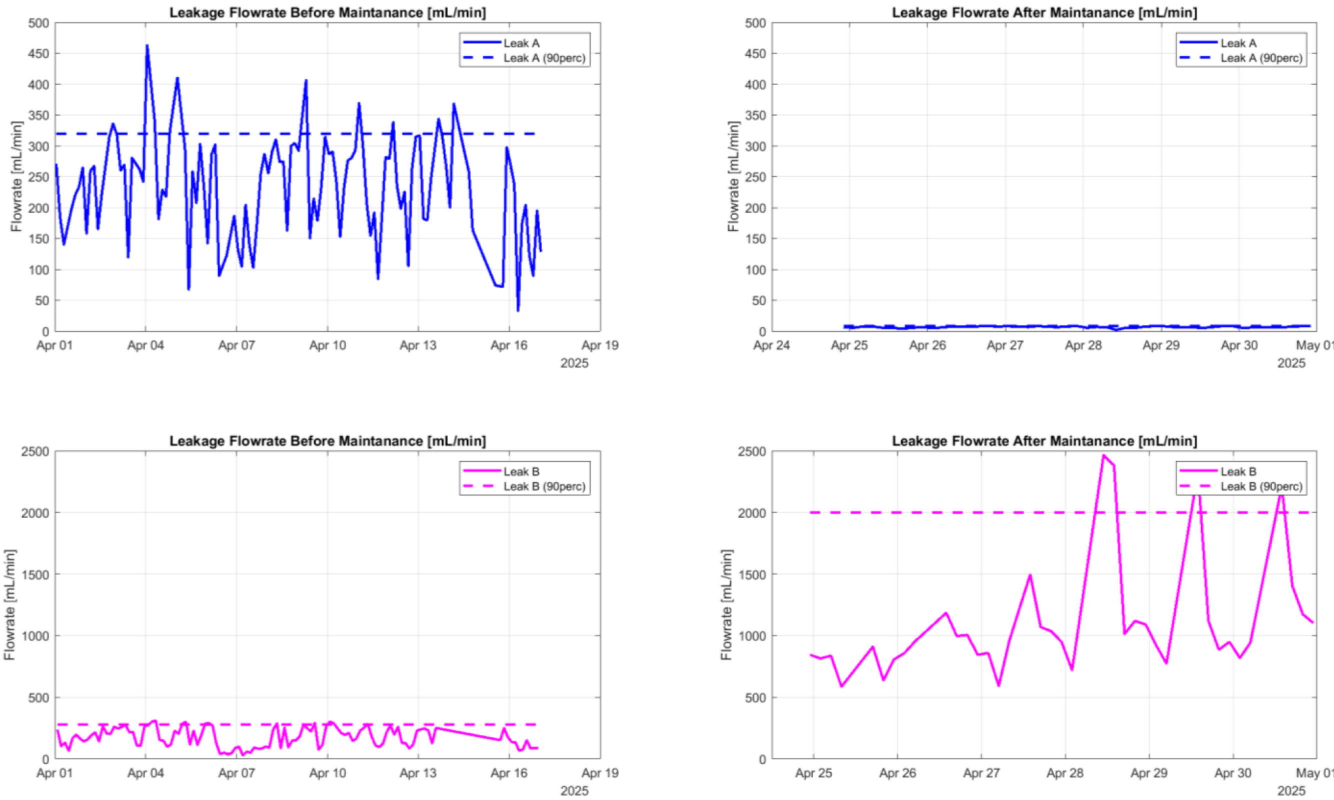

Figure 6—Time-series data from the GSN sensor showing the methane concentration trend before and after leak repair for two monitored leaks: (up) the Leak A; (bottom) the Leak B. The drop to baseline levels confirms successful mitigation, while an increase of flowrate level triggered reinspection.

These key events are summarized in Tab. 2, which provides an overview of detection timing, verification methods, corrective actions, and post-maintenance outcomes for each leak.

Table 2—Summary of field leak events and verification outcomes.

Leak	Detection Time	Verification Method	Action taken	Outcomes
Leak A	Pre-existing; detected within few hours after GSN installation	Targeted inspection with portable instruments, including HFS and FID	Gasket replaced and flange tightened.	Leak resolved; sensor reading dropped post-maintenance.
Leak B	Pre-existing; detected within few hours after GSN installation	Targeted inspection with portable instruments, including HFS and FID	Gasket replaced and flange tightened.	Leak rate spiked after first fix; second repair attempt triggered.
Leak C	Pre-existing; detected within few hours after GSN installation	Targeted inspection with portable instruments, including HFS and FID	No action taken	Smaller leak, confirmed.

The pilot also demonstrated the practical durability and coverage of the GSN technology under real field conditions. The sensors were exposed to challenging environmental factors such as salty marine air, strong winds, heavy solar exposure or rain. The hardware proved robust: the IP66-rated enclosures prevented water ingress even during downpours, and no sensor failures were recorded. The solar panels continued to recharge the supercapacitor power modules adequately, even in the presence of salt film accumulation, which was easily removed during a routine maintenance. The power system ensured continuous node operation through multiple overcast days without downtime. Moreover, the long-range LoRa wireless network remained reliable across complex industrial environments, including facilities with dense steel infrastructures. All installed nodes consistently communicated with the on-site gateway, which achieved 100% uptime. Overall data completeness remained around 97%, with minor gaps mainly due to temporarily shaded nodes – later resolved through panel repositioning. These results confirm that the GSN system can be deployed at scale in diverse industrial settings with minimal maintenance, offering strong resilience to challenging environmental conditions.

The field implementation across three distinct sites provided a compelling case for continuous methane monitoring. On one hand, it gave assurance where things were in good shape: a site with effectively no leaks was able to prove its integrity continuously (something periodic surveys alone cannot easily do), giving operators peace of mind and potentially allowing optimization of survey frequency. On the other hand, the system delivered an early warning and a detailed diagnosis of leaks that enabled swift corrective action – preventing ongoing losses and emissions that might have otherwise continued for months. This continuous record, complete with location-specific insights, transforms leak detection and repair into a proactive and verifiable process. In summary, the multi-site GSN pilot demonstrated that deploying a dense network of low-power methane sensors is feasible and highly effective for real-time LDAR: it reliably detects even minute leaks, pinpoints and quantifies significant emission sources for fast response, and provides ongoing confirmation of a facility's emission status with fine detail.

Results

The pilot deployments confirmed that the GSN system delivers reliable, high-quality data. Thanks to local buffering and automatic retransmission, the network maintained over 99% data completeness and near-continuous uptime (97–100%), ensuring trustworthy monitoring. The system estimates methane leak rates above ~1 g/h with ±65% accuracy. Over time, site-specific calibration can further improve accuracy, using observed ppm drops after repairs.

The pilots showed the clear value of continuous monitoring. GSN identified hidden or intermittent leaks early, enabling rapid repair and confirming the results in real time. One ongoing leak, emitting up to 0.013 kg/h, was fixed within days—preventing up to 0.11 tons/year of methane emissions. In another case, a leak was detected in its incipient phase, allowing for a prompt and proactive intervention before it could evolve into a more significant issue. The system also confirmed the absence of emissions in unaffected areas, supporting effective maintenance planning and operational awareness.

This minimum detection capability is below the threshold prescribed by EU Regulation 2024/1787 (<1 g/h), providing sensitivity well beyond current regulatory requirements. The system also proved effective in reconciling L5 data for a plant section in alignment with OGMP gold-standard methodologies, thus demonstrating suitability for high-tier reporting.

A reconciliation exercise, compliant to OGMP 2.0, has been conducted comparing mass flow estimates, referred to a plant section, obtained through three different methods: GSN sensor quantification, drone-based measurements, and OGI camera by applying "API no-leak emission factors methodology". According to Company standards the OGMP Level 4 implementation plan foreseen sampling ~5-20% of the fugitive emission sources depending on the population size and emission materiality. Selected plant section has 770 total possible fugitive emission sources, in the same area GSNs were installed in 53 possible fugitive emission sources equivalent to 6,88%. The results collected in this sample were then extended to the entire section. As shown in Fig. 7, the GSN-estimated flow rates for the identified emission sources were consistent with drone measurements, with overlapping uncertainty intervals. The OGI values were systematically lower as technique is way less sensitive and cannot identify leaks detected by GSN. This comparison confirms the robustness of the GSN estimates over time and under variable operational conditions. The reconciliation supports the conclusion that GSN data can be used not only for early leak detection and trend analysis but also as a credible source for quantifying emissions in compliance with methane reporting frameworks. In this context, "compliant" indicates that the GSN measurement falls within the confidence interval of the reference method, meaning that the two estimates are statistically consistent and belong to the same distribution. This level of agreement supports the acceptability of GSN quantification for operational and reporting purposes.

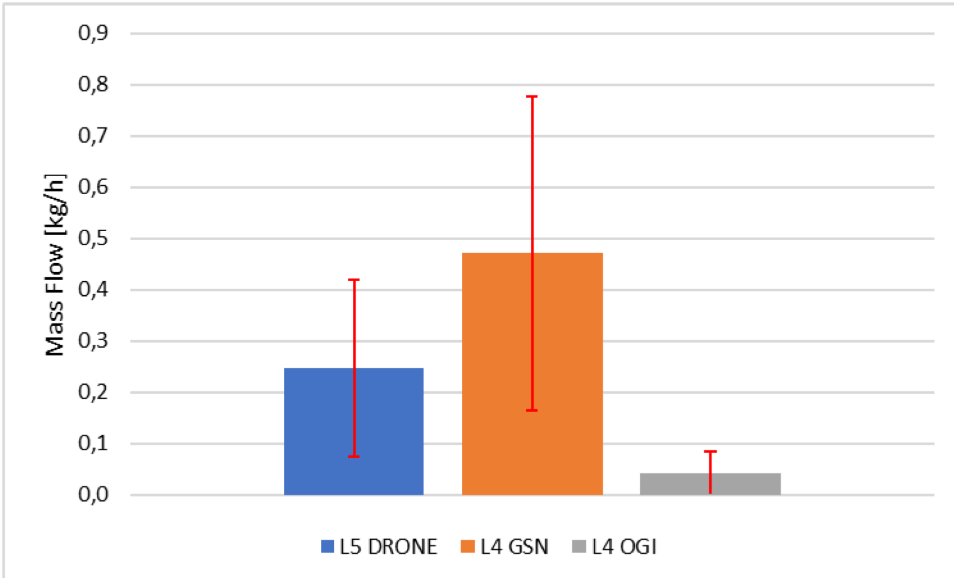

Figure 7—Comparison of methane emission estimates obtained through three independent methods (GSN sensor data, drone-based quantification, and OGI Camera).

The GSN system introduces several key improvements over conventional leak detection and repair (LDAR) practices, based on our observations:

1. Continuous 24/7 Monitoring vs. Intermittent Surveys: GSN ensures 24/7 coverage of potential leak points, eliminating the temporal gaps of manual LDAR surveys. During the pilots, the system detected transient and early-stage leaks that would have likely been missed between inspections. Leak detection becomes continuous and automated, allowing for rapid intervention.
2. Source-Specific Quantification of Leaks: Unlike threshold-based detectors, GSN nodes estimate methane flow rates at each source, enabling prioritization by severity and empirical calculation of total emissions. This transforms leak management into a measurable, data-driven process that supports both mitigation and reporting.
3. Reduced Human Exposure and Operational Effort: By automating routine surveillance, GSN reduces the need for personnel to inspect high-risk or hard-to-reach areas. This improves safety and frees up resources to focus on confirmed issues. Over time, the frequency of manual LDAR activities can be optimized, reducing costs and disruptions.
4. Rapid Response and Maintenance Integration: GSN enables immediate response to leaks via real-time notification, preventing escalation and improving asset integrity. Data trends support condition-based maintenance, and the system can integrate with digital work-order platforms. Local edge logic ensures fast detection, while cloud analytics support deeper insight and planning.
5. Scalability and Deployment Flexibility: Wireless, low-power GSN nodes are easy to install and scale across large facilities. The dense sensor coverage achieved in the pilots revealed emission hotspots with precision, guiding targeted repairs and long-term improvements. Unlike conventional detectors, GSN enables granular and facility-wide leak surveillance.

Overall, the pilot campaigns demonstrated that the GSN system enables a step change in methane leak management. By combining high-fidelity data, continuous monitoring, and source-specific quantification, the network delivers faster detection, smarter prioritization, and safer, more efficient operations. The integration of real-time notifications, scalable deployment, and actionable insights position GSN as a powerful tool for transforming leak detection from a periodic obligation into a continuous, proactive process —ready to meet both operational needs and emerging regulatory expectations.

Conclusion

The pilot deployments of the Gas Sensor Node (GSN) system have demonstrated that continuous, source-level methane monitoring is not only technically feasible—it represents a novel and scalable breakthrough in emissions management. Unlike traditional leak detection approaches, GSN brings automation, real-time quantification, and integration with digital workflows directly to the field. This innovation transforms methane monitoring from a reactive task into a continuous, intelligence-driven process, unlocking both operational and strategic value.

Key strategic benefits include:

- OGMP 2.0 Compliance

  The GSN network supports full alignment with the Oil and Gas Methane Partnership 2.0 framework, enabling direct, source-level measurements in line with OGMP Level 4. Rather than relying on emission factors or occasional surveys, operators can now quantify actual leak rates from individual components in real time. All data is time-stamped and archived, simplifying annual reporting and providing verifiable proof of emissions reductions. During implementation campaign, the system measured an emission for the considered plant section of 0,471±0,304 kg/h —consistent with what a site level verification method (e.g., drone or satellite) would expect—supporting the OGMP Level 5 "gold standard" reconciliation.
- Regulatory Readiness:

  With methane regulations tightening globally (e.g., the upcoming EU Methane Regulation), GSN positions adopters ahead of the compliance curve. Its continuous monitoring capabilities may satisfy or reduce the need for manual LDAR surveys, where permitted. Early implementation demonstrates regulatory responsibility and future-proofs operations without the need for major retrofits, giving the site a proactive "regulation-ready" posture.
- Progress toward Net Zero and ESG Goals:

  GSN directly supports corporate climate strategies and ESG reporting. In the site where the three significant leaks were recorded, real-time data enabled repairs that are now preventing approximately 0.25 tons/year of methane emissions—significant for a single site. Across multiple facilities, this impact scales substantially. Beyond mitigation, GSN provides credible, measurement-based evidence of progress, responding to increasing scrutiny from investors and stakeholders demanding transparency and accountability in emissions reporting.
- Digital Transformation and Operational Excellence:

  The system exemplifies how Industrial IoT can enhance not just compliance, but overall plant intelligence. Real-time data feeds into digital dashboards and can support machine learning models or digital twins. Over time, this enables predictive maintenance, reduces unplanned downtime, and shifts leak detection from an HSE afterthought to an integrated element of routine operations. The workforce, too, evolves—empowered by visibility and data-driven culture.
- Industry Leadership and Complementary Use of Technologies:

  Deploying GSN positions the company at the forefront of methane management innovation. It sets a benchmark for best practices while complementing other technologies like satellites, drones, and handheld detectors. GSN fills the critical ground-level gap—providing persistent coverage and immediate alerts inside facilities—within a multi-layered monitoring strategy. This integrated ecosystem of tools ensures both broad and granular oversight, aligning with expert recommendations for comprehensive emissions control.

In summary, the Gas Sensor Node system introduces a fundamentally new way to manage methane emissions—combining innovation, automation, and digital integration. The pilot results confirm its potential to scale across facilities, enabling faster interventions, more accurate reporting, and measurable environmental benefits. As the oil and gas industry moves toward net-zero and increased accountability, technologies like GSN offer not just compliance—but leadership.

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